SC2104 – SENSORS, INTERFACING AND DIGITAL CONTROL

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PART II – INTRODUCTION TO CONTROL THEORY/SYSTEM

Lecture 1 & 2

CONTROL THEORY IN ACTION

QUADCOPTERS (TED TALK 2013)

The astounding athletic power of quadcopters *Raffaello D'Andrea*

https://www.youtube.com/watch?v=w2itwFJCgFQ

DOUBLE INVERTED PENDULUM

Video\Control of Double Inverted Pendulum, WETI Gdańsk_2.mp4

OBJECTIVE OF THIS MODULE

- Why control for computer engineers
- To understand analogue systems
- To understand discrete-time systems
- To be able to analyze the system characteristics
- To be able to design sampled-data controllers

REFERENCE BOOKS

- Text Book:
  - Benjamin C. Kuo, Digital Control System, Oxford University Press, Second Edition, USA, 1995
  - Tim Wescott, Applied Control Theory for Embedded Systems, 2006
- Reference Books:
  - Katsuhiko Ogata, Discrete-time control systems, Prentice Hall, 2 Edition, 1995.
  - Åström K.J and Wittenmark B.: *Computer Controlled Systems* 3ed. *Prentice* Hall, 1997.

NOTE ON LECTURE/LECTURER STYLE

- Each lecture will last about 50 x 2 minutes.
- Do ask questions during the Lecture or during the Break.
- You can also send your questions by email or meet me in the office, if many students have similar doubt, I will provide additional examples/clarifications in the next class.
- Any update on lecture notes will be uploaded onto NTULearn.
- IMPORTANT: **EXAM** are Compulsory.

CONTENT OF THIS MODULE

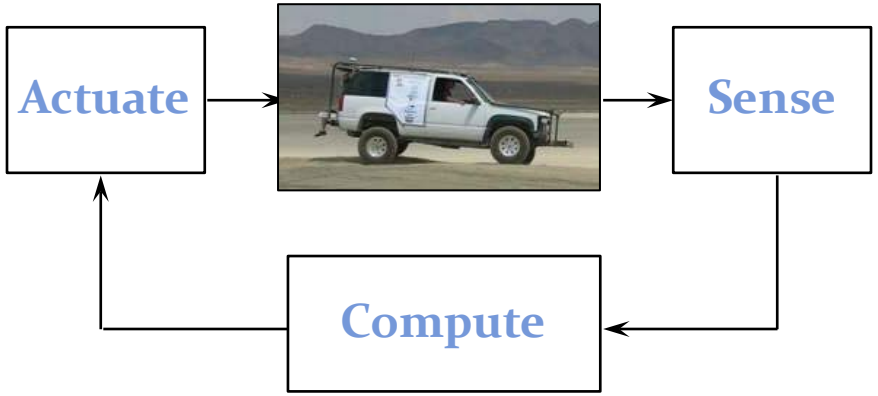
- Lecture 1 – Introduction to Control Systems
- Lecture 2 - Analogue Control System
- Lecture 3 – Laplace Transform, Transfer function,
- Lecture 4 - Analysis in S-Plane
- Lecture 5 – Digital Control System
- Lecture 6 – z-Transform, Transfer function
- Lecture 7 – Analysis in z-Plane
- Lecture 8 – Stability
- Lecture 9 – Basic Controller Design
- Lecture 10 – Implementation of a Digital Controller

MOTIVATION FOR CONTROL ENGINEERING

- History of feedback control started when human desire to harness the materials and forces of nature to their advantage
- Early examples for control devices
  - clock regulating systems
  - mechanisms for keeping wind-mills pointed into the wind.
- Modern industrial plants have sophisticated control systems which are crucial to their successful operation.
- Control Engineering has had a major impact on society.

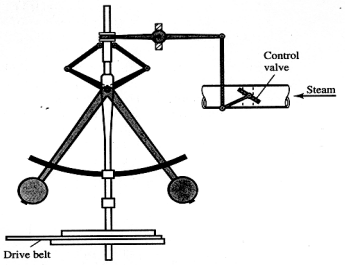
EXAMPLES

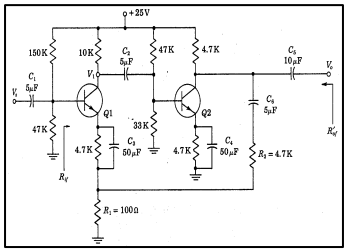
Control = Sensing + Computation + Actuation

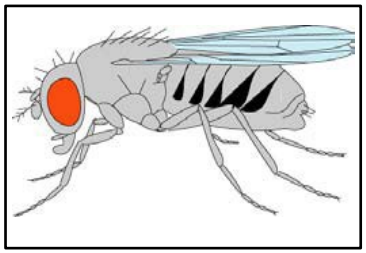


- Feedback Principles
  - Robustness to Uncertainty
  - Design of Dynamics

Many examples of feedback and control in natural & engineered systems:





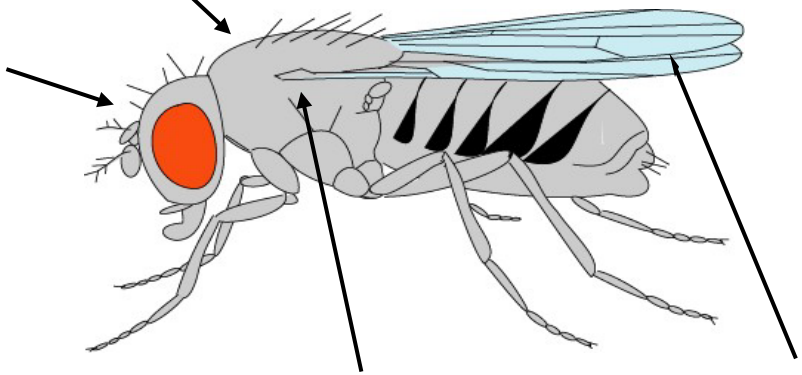


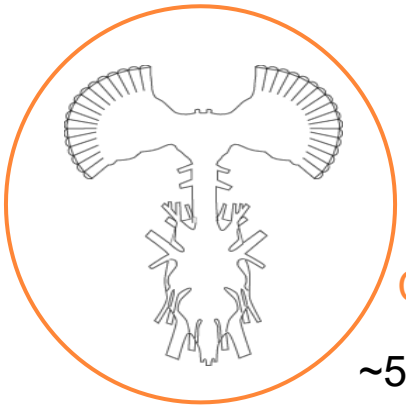
NATURAL CONTROL SYSTEM – INSECT FLIGHT

SENSING

hind wing gyroscopes (halteres)

neural superposition eyes





COMPUTATION

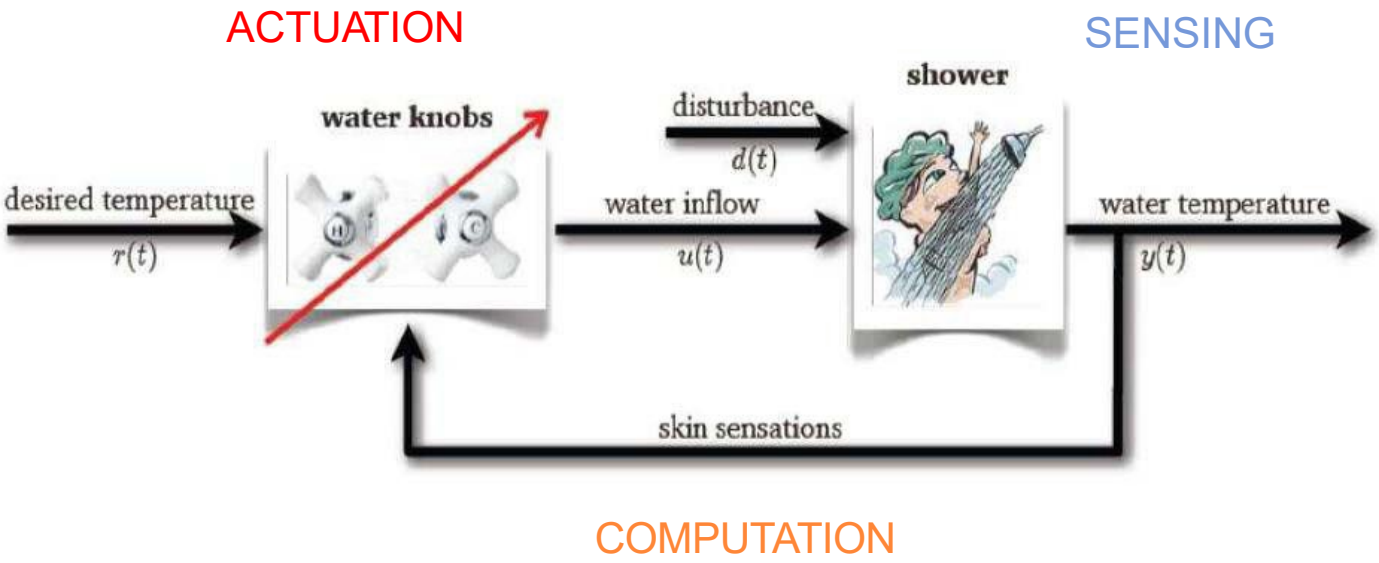
~500,000 neurons

specialized "power" muscles

two wings (di-ptera)

ACTUATION

NATURAL CONTROL SYSTEM – HUMAN IN SHOWER

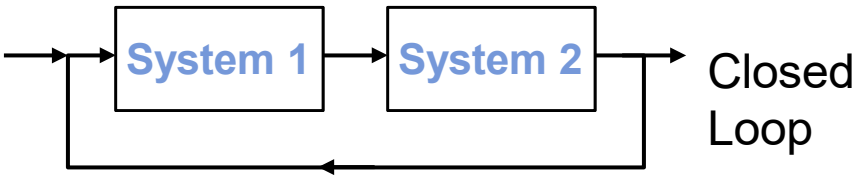


- Water flow $u(t)$ between hot and cold knobs need to be controlled
- Skin sensors measures tolerable temperature $y(t)$
- Temperature fluctuations $d(t)$ due to atmosphere, mood and etc

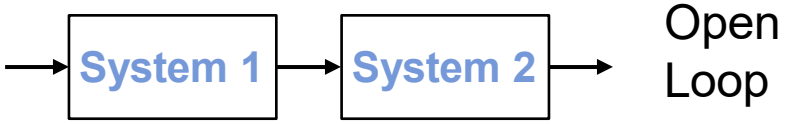
WHAT IS FEEDBACK?

- Definition of feedback as in Miriam Webster:
  - the return to the input of a part of the output of a machine, system, or process (as for producing changes in an electronic circuit that improve performance or in an automatic control device that provide self-corrective action) [1920]

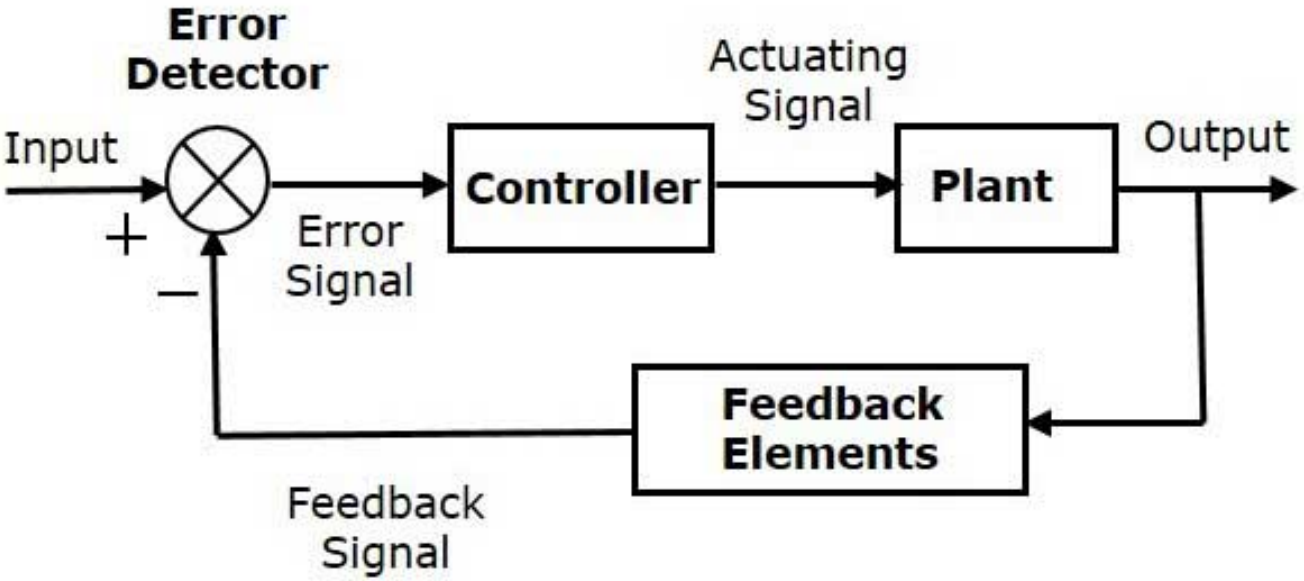
- Feedback = mutual interconnection of two (or more) systems
  - System 1 affects system 2
  - System 2 affects system 1
  - Cause and effect is tricky; systems are mutually dependent



- Feedback is ubiquitous in natural and engineered systems

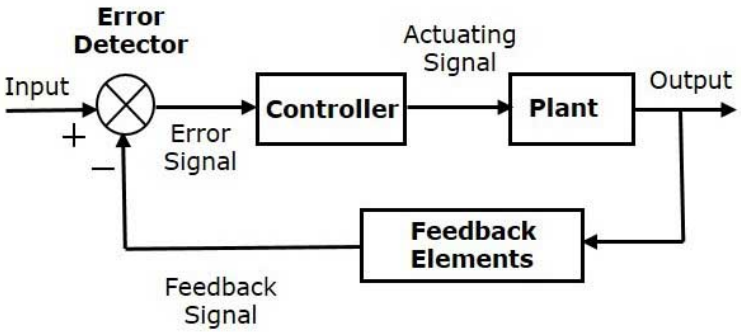


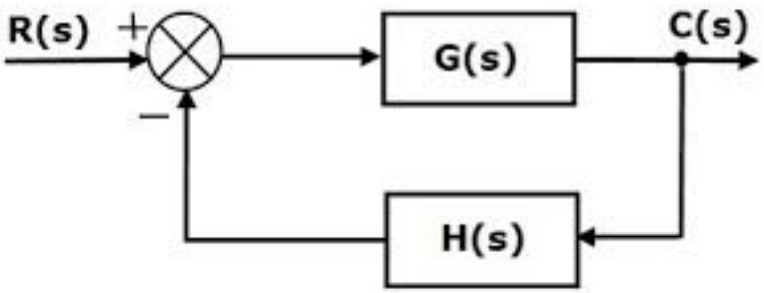
BLOCK DIAGRAM OF A CONTROL SYSTEM



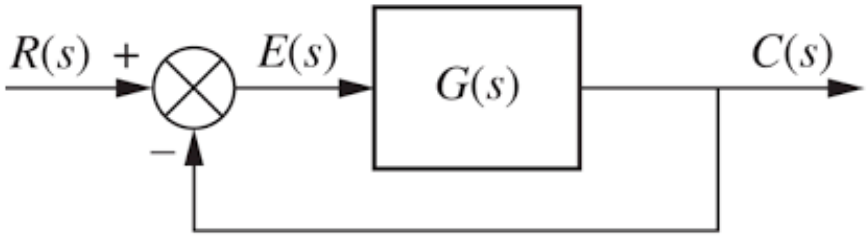
- Objective
  - System **output** follows **reference** as close as possible without **human interference**, i.e., make **error** as small as possible.
- Key issues:
  - How to describe the system to be controlled? (Modeling)
  - How to design the controller?(Control)

BLOCK DIAGRAM OF A CONTROL SYSTEM



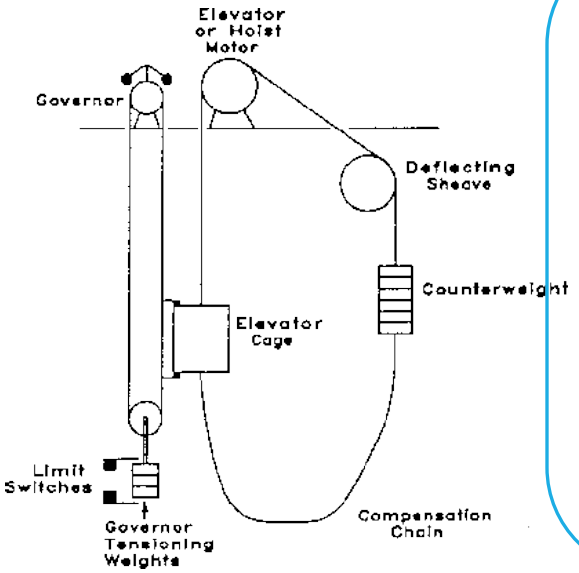


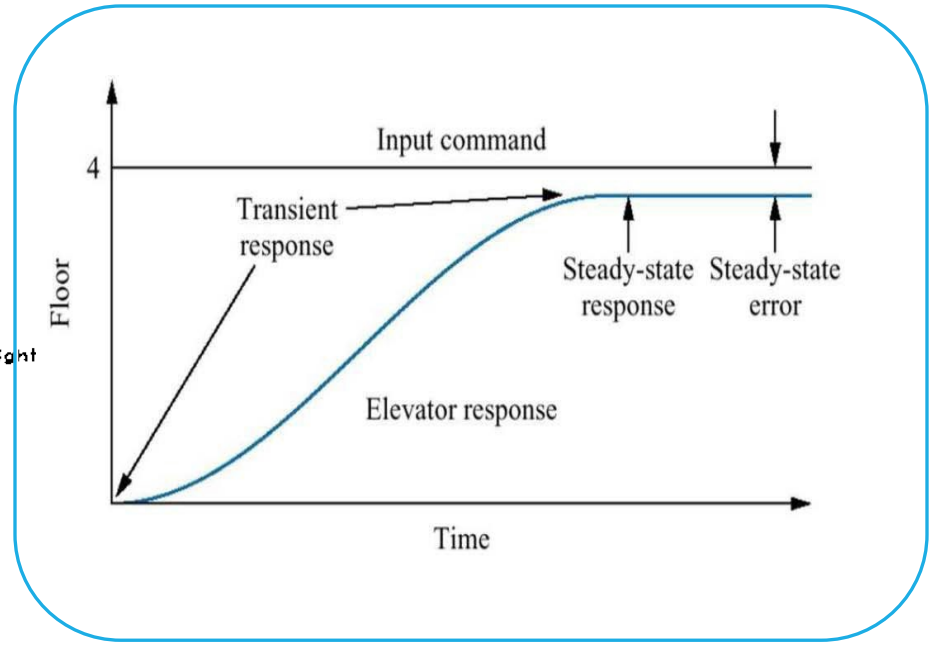
Negative Feedback Control System



Unity Gain Negative Feedback Control System

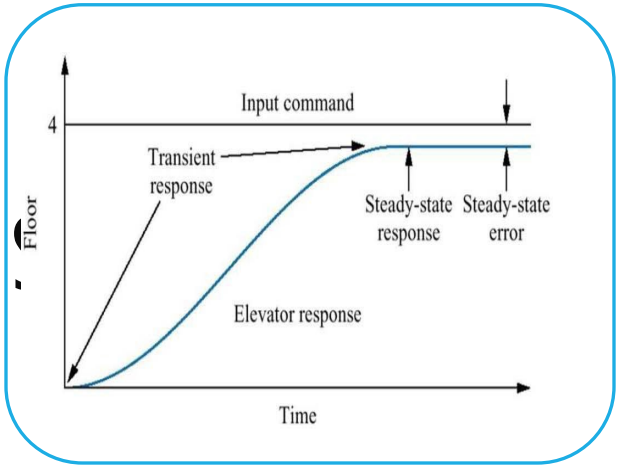
RESPONSE CHARACTERISTIC OF A LIFT CONTROL SYSTEM





- Reference input: Go to level 4
- Response:
  - Lift/Elevator rises gradually to level four. This is called the **transient response**.
  - Steady-state response: It is an approximation to desired input
  - Final response: elevator may not stop exactly in level four. The difference is called **steady-state error**.

RESPONSE CHARACTERISTIC OF A LIFT CONTROL

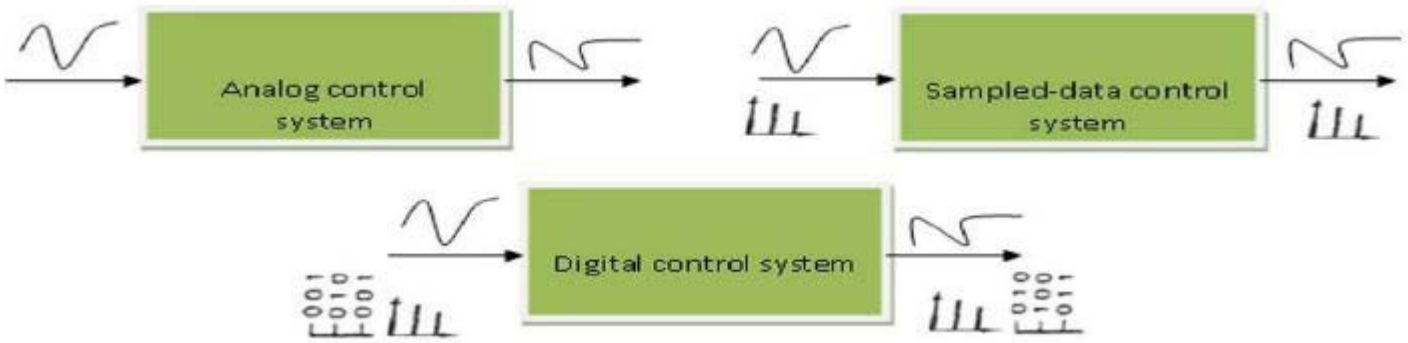


- System respond to reference input and undergo a transient response before reaching steady-state response that generally resembles the input.
  - **Transient response**:
    - Slow response makes passengers impatient
    - Excessively rapid makes it uncomfortable
    - Oscillation after arrival makes it uncomfortable
    - One need to adjust the behavior such that it is comfortable for passengers
  - **Steady-state response**:
    - Level exactly with the floor such that steady-state error is minimum.
  - **Stability**:
    - Actual response of elevator is the sum of natural response and forced response (external input). If natural response grows without bound the system no longer controlled or unstable, i.e., lift always crash to the floor or exit through ceiling. Control system should be designed to be stable.

TYPICAL STEPS IN CONTROL SYSTEM DESIGN

- Establish Control Objectives
  - Qualitative (reduce fuel) and Quantitative ($T_s < 5$sec) requirements
  - Understand the process (expected demand and disturbance) and goals
  - Strong Understanding of the system dynamics – to avoid inappropriate goals
- Select Sensors and Actuators
  - What to be sensed and controlled ?
  - Cost, reliability and size
- Mathematical Model of Process
  - Obtain model for controller evaluation (accuracy!!)
- Design Controller
  - Choose a method and select parameters
- Analyze Closed-loop Performance
  - Does it meet the objectives?

CONTROL SYSTEM TYPES



A. Analog control system or Continuous-time control system
   - Signals are continuous-time

B. Sample data control system
   - Signals include both sampled data and continuous-time

C. Digital control system
   - Signals include digital, sampled-data and continuous-time

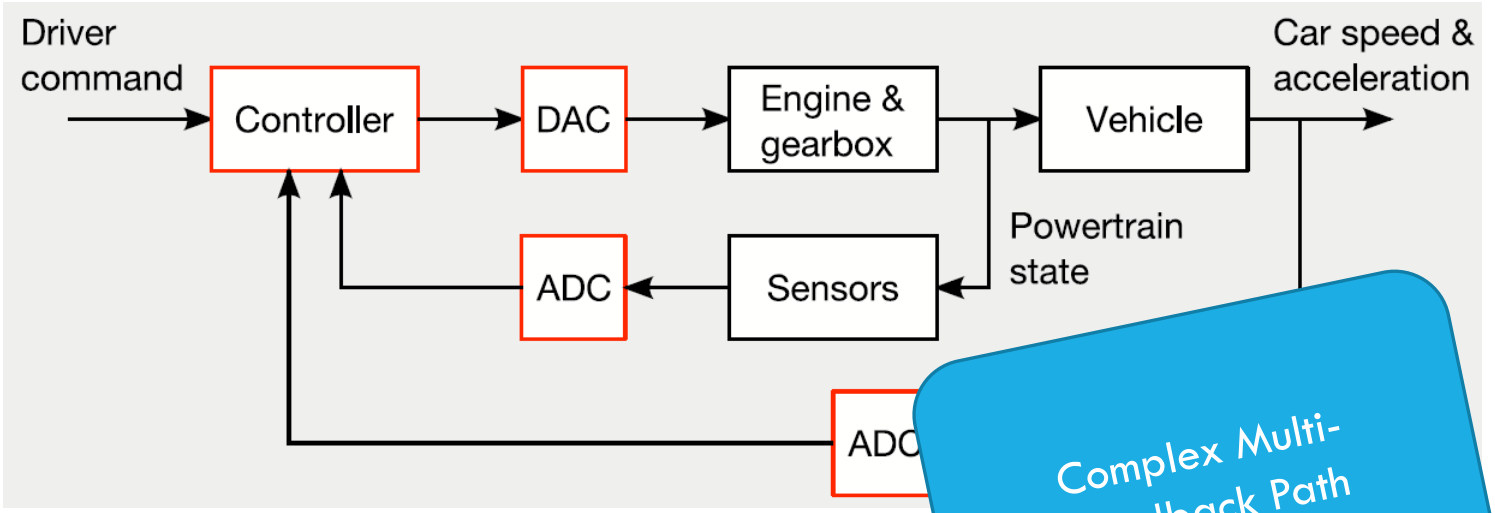
WHY DIGITAL CONTROL SYSTEM?

- Accuracy
  - Digital signals are usually represented using binary patters (12 bits or more)
- Reliability
  - Processing digital signals avoid noise and uncertainty that affects analogue signal processing
- Flexibility
  - Limited only by processing power (also programming ability)
- Cost
  - Advances in embedded controller make digital controller small and cost effective.

DISADVANTAGE OF DIGITAL CONTROL

- Mathematical analysis and design of discrete-time control system is complex and tedious
- Delay due to Analog-to-Digital and Digital-to-Analog conversion (non-zero sampling time) influence the performance significantly.
- Design with these delay is difficult.
- Quantization error affects the accuracy of digital control.

CONTROL SYSTEM IN AUTOMOBILE



- At each sampling instant
  - Reads in sensor measurements from ADCs
  - Uses these to compute the required control inputs
  - Outputs control signals to actuators via DACs.

CONTROL THEORY IN COMPUTING SYSTEMS

- Network flow controllers (TCP/IP – RED)
- Admission control
- QoS in Caching
- Apache QoS differentiation
- CPU utilization control in cloud
- A self-tuning memory management of a database system

SUMMARY – INTRODUCTION TO CONTROL SYSTEMS

- Examples
- What is feedback?
- Control system = sensors + control algorithm+ actuators
- Feedback control System & approaches

